

Exercise (Alternative definition for semantic security). *The standard notion of semantic security is defined through the following games:*

$$\begin{array}{cc}
 \mathcal{G}_0^{\mathcal{A}} & \mathcal{G}_1^{\mathcal{A}} \\
 \left[\begin{array}{l} \text{sk} \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}^{\text{Enc}_{\text{sk}}(\cdot)} \\ m \leftarrow \mathcal{M}_0 \\ c \leftarrow \text{Enc}_{\text{sk}}(m) \\ \text{return } [g(m) \stackrel{?}{=} \mathcal{A}(c)] \end{array} \right. & \left[\begin{array}{l} \text{sk} \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}^{\text{Enc}_{\text{sk}}(\cdot)} \\ m \leftarrow \mathcal{M}_0, \bar{m} \leftarrow \mathcal{M}_0 \\ \bar{c} \leftarrow \text{Enc}_{\text{sk}}(\bar{m}) \\ \text{return } [g(m) \stackrel{?}{=} \mathcal{A}(\bar{c})] \end{array} \right.
 \end{array}$$

where the second game $\mathcal{G}_1$ models a very specific attack in the setting where the adversary does not see the encryption of a challenge message. This does not reflect reality close enough as the adversary can perform other more successful attacks in this setting. To capture that we define a new security game

$$\mathcal{G}_2^{\mathcal{A}_*} \left[\begin{array}{l} \text{sk} \leftarrow \text{Gen} \\ \mathcal{M}_0 \leftarrow \mathcal{A}_*^{\text{Enc}_{\text{sk}}(\cdot)} \\ m \leftarrow \mathcal{M}_0 \\ \text{return } [g(m) \stackrel{?}{=} \mathcal{A}_*] \end{array} \right.$$

This allows us to define two advantages

$$\begin{aligned}
 \text{Adv}_g^{\text{sem}}(\mathcal{A}) &= \Pr [\mathcal{G}_0^{\mathcal{A}} = 1] - \Pr [\mathcal{G}_1^{\mathcal{A}} = 1] \\
 \text{Adv}_g^{\text{sem}*}(\mathcal{A}, \mathcal{A}_*) &= \Pr [\mathcal{G}_0^{\mathcal{A}} = 1] - \Pr [\mathcal{G}_2^{\mathcal{A}_*} = 1]
 \end{aligned}$$

The cryptosystem is (t, ε) -weakly semantically secure if for any t -time adversaries $\mathcal{A}$ and $\mathcal{A}_*$ the advantage $\text{Adv}_g^{\text{sem}*}(\mathcal{A}, \mathcal{A}_*) \leq \varepsilon$. Prove that semantic security implies weak semantic security for the same function g . Prove that weak semantic security implies semantic security. Interpret the results.

Solution.